

Rupa Jhade

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PROFESSIONAL SUMMARY

Data Engineer with 4+ years of experience owning and operating large-scale, production data pipelines across banking, insurance, healthcare analytics, education, and manufacturing domains in regulated enterprise environments. Strong background in building resilient Spark and Airflow systems on cloud platforms, enforcing data quality and governance standards, and delivering reliable analytics for business-critical decision-making. Known for end-to-end ownership, operational excellence, and applying AI-driven monitoring and governed LLM solutions to improve data reliability at scale.

EXPERIENCE

EXL Services

May 2024 - Present

Data Engineer

Pittsburgh, PA

- **Data Pipeline Development:** Designed and supported scalable data pipelines using PySpark, Spark SQL, and Azure Databricks for high-volume banking transactions and customer data, enabling reliable analytics and reporting.
- **ETL Data Ingestion:** Built ETL workflows ingesting data from AWS S3, RDS, APIs, and Azure Data Factory, transforming raw banking data, including JSON API sources, into structured, analytics-ready datasets for compliance and business reporting.
- **Workflow Automation:** Automated batch data pipelines using Apache Airflow, implementing task dependencies, retries, and monitoring to ensure SLA-driven data delivery.
- **Performance Tuning:** Optimized Spark jobs using partitioning, Parquet file formats, and broadcast joins, reducing data processing time by 25–30%.
- **Data Validation:** Implemented data validation checks including null handling, duplicate detection, and schema enforcement, improving reliability of financial datasets.
- **Data Security Compliance:** Assisted with access controls and data masking to ensure secure handling of sensitive banking data according to security and governance standards in hybrid cloud environments.
- **CI/CD Integration:** Supported CI/CD-aligned deployment of data pipelines using Git-based versioning, Dockerized jobs, and controlled promotion across environments to ensure reliable releases.
- **Team Collaboration:** Partnered with cross-functional teams, including client analysts, BI developers, and Azure admins, to refine data models and support UAT for banking dashboards and reports.

Rochester Institute of Technology - GCCIS

Aug 2023 - Apr 2024

Data Engineer

Rochester, NY

- **Data Integration:** Designed and developed end-to-end data pipelines using SQL, PySpark, and Azure Data Factory to integrate academic, co-op, and financial data from multiple systems, improving data freshness by 80% and enabling near real-time analytics.
- **Data Modeling:** Built scalable data models and optimized schema design to support reporting and predictive analytics, ensuring consistency, accuracy, and data reliability across institutional and cloud-based databases.
- **Workflow Orchestration:** Automated ingestion and transformation processes using Airflow, Docker, and Git, improving pipeline reproducibility and deployment efficiency by 40%.
- **Performance Optimization:** Implemented data validation, partitioning, and indexing strategies to enhance query speed and reduce data latency by 30%.
- **Data Governance:** Collaborated with IT and institutional research teams to establish governance and validation frameworks aligned with industry security standards, reducing manual reconciliation and ensuring cross-departmental data accuracy.
- **Reporting & Dashboards:** Developed interactive Tableau dashboards to monitor key pipeline metrics such as data latency, throughput, and load success rate, enabling proactive issue detection and executive decision-making transparency.
- **Documentation & Collaboration:** Created reproducible workflows, maintained version-controlled repositories, and prepared detailed technical documentation to standardize analytics and engineering practices.

Sree Rayalaseema Hi-Strength Hypo Ltd.

Jun 2022 - Apr 2023

Data Engineer

Hyderabad, India

- **ETL Development:** Built automated Python and SQL ETL pipelines integrating DCS and laboratory data, reducing manual preparation time by 30% and improving data reliability.
- **Process Optimization:** Implemented Statistical Process Control (SPC) and conducted hypothesis testing to monitor chemical process stability, reducing process rework by 15%.
- **Operational Efficiency:** Applied optimization algorithms and automated dashboards to balance resource usage, increase throughput, and reduce energy costs.

- **Automation & Reporting:** Established validation checks and data reconciliation processes, ensuring accuracy across production and inventory systems.
- **Production Data Support:** Worked with process engineers and operations teams to translate analytical insights into actionable improvements, driving data-informed decisions.

Accenture

Jan 2021 - May 2022

Data Analyst

Hyderabad, India

- **Data Analysis:** Conducted comprehensive analysis of employee performance, recruitment, and attrition trends using Python, SQL, and Power BI, uncovering actionable insights that supported HR strategic initiatives.
- **Dashboarding:** Developed interactive Power BI dashboards visualizing attrition, recruitment, and diversity KPIs, reducing manual reporting time by 35%.
- **Cloud & Automation:** Supported migration of legacy HR reports to AWS cloud and automated recurring workflows with Python, automating recurring workflows, saving 25% manual effort.
- **Data Governance:** Created standardized data definitions and consistency checks across HR data pipelines, ensuring compliance with corporate reporting standards.
- **Collaboration & Delivery:** Coordinated with HR business partners and IT stakeholders to deliver data-driven dashboards supporting strategic workforce planning.
- **Impact:** Delivered an end-to-end HR analytics solution enabling real-time workforce reporting and data-driven retention strategies.

PROJECTS

AI-Driven Data Quality & Observability Platform

- Built an AI-enabled data observability platform using PySpark and Apache Airflow to detect data quality issues such as null spikes, schema drift, duplicates, and volume anomalies in batch pipelines, applying ML-based anomaly detection and LLM-powered root cause analysis with automated alerting and dashboards, improving issue detection time by approximately 30% and enhancing overall pipeline reliability

Secure LLM-Powered Conversational Analytics Platform

- Developed a secure LLM-based conversational analytics system enabling natural language access to enterprise databases, implementing SQL validation, guardrails, and role-based access controls with optimized query generation and automated visual summaries, reducing manual and ad-hoc reporting requests by approximately 25% while maintaining data governance

TECHNICAL SKILLS

- **Azure & Cloud Platforms:** Azure Data Factory, Azure Databricks, Azure SQL Database, Azure Synapse Analytics, AWS (EC2, S3, Lambda, RDS, Redshift, Glue, Kinesis), GCP, Snowflake
- **Data Engineering & Architecture:** ETL Development, Data Modeling, Schema Design, Pipeline Automation, Data Integration, Data Warehousing, Data Governance, Data Validation, Deduplication, Dimensional Modeling
- **Visualization & Tools:** Tableau, Power BI, Power Query, Splunk, Matplotlib, Seaborn, DAX, Streamlit, Flask, MLflow, Advanced Excel
- **AI / ML:** Anomaly Detection, Machine Learning, Explainable AI, Model Evaluation
- **GenAI / LLMs:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Natural Language to SQL (NL2SQL), Secure LLM Workflows, Prompt Engineering for Structured Outputs
- **APIs & Data Integration:** API Data Ingestion, OAuth 2.0, JSON Normalization, Docker, Git, Linux
- **Programming & Big Data:** Python, SQL, PySpark, Spark, Hadoop, Airflow, Flink, NoSQL, PostgreSQL, Hive, Partitioning Strategies, Parquet

EDUCATION

Rochester Institute of Technology

Master of Science, Data Science

Rochester, NY

- **GPA:** 3.6

Amrita Vishwa Vidyapeetham

Bachelor of Technology, Computer Science and Engineering

Coimbatore, India

- **GPA:** 4.0

CERTIFICATIONS

- AWS Certified Data Engineer – Associate
- Snowflake Data Engineer Certification
- IBM Data Science Professional Certificate